efficiency. Other works [10], [101] apply self-attention mechanisms for channel & spatial attention to explore pairwise interaction. Yet further works [120], [124] adopt the spatial-channel attention mechanism to enlarge the receptive field.

Representative channel & spatial attention mechanisms and specific process $g(x)$ and $f(g(x), x)$ described as Eq. 1 are in given Tab. 7; we next discuss various ones in detail.

3.6.1 Residual Attention Network

Inspired by the success of ResNet [145], Wang et al. [119] proposed the very deep convolutional *residual attention network* (RAN) by combining an attention mechanism with residual connections.

Each attention module stacked in a residual attention network can be divided into a mask branch and a trunk branch. The trunk branch processes features, and can be implemented by any state-of-the-art structure including a pre-activation residual unit and an inception block. The mask branch uses a bottom-up top-down structure to learn a mask of the same size that softly weights output features from the trunk branch. A sigmoid layer normalizes the output to $[0, 1]$ after two 1×1 convolution layers. Overall the residual attention mechanism can be written as

$$s = \sigma(\text{Conv}_2^{1\times1}(\text{Conv}_1^{1\times1}(h_{\text{up}}(h_{\text{down}}(X))))) \tag{78}$$

$$X_{out} = sf(X) + f(X) \tag{79}$$

where h_{up} is a bottom-up structure, using max-pooling several times after residual units to increase the receptive field, while h_{down} is the top-down part using linear interpolation to keep the output size the same as the input feature map. There are also skip-connections between the two parts, which are omitted from the formulation. f represents the trunk branch which can be any state-of-the-art structure.

Inside each attention module, a bottom-up top-down feedforward structure models both spatial and cross-channel dependencies, leading to a consistent performance improvement. Residual attention can be incorporated into any deep network structure in an end-to-end training fashion. However, the proposed bottom-up top-down structure fails to leverage global spatial information. Furthermore, directly predicting a 3D attention map has high computational cost.

3.6.2 CBAM

To enhance informative channels as well as important regions, Woo et al. [6] proposed the *convolutional block attention module* (CBAM) which stacks channel attention and spatial attention in series. It decouples the channel attention map and spatial attention map for computational efficiency, and leverages spatial global information by introducing global pooling.

CBAM has two sequential sub-modules, channel and spatial. Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$ it sequentially infers a 1D channel attention vector $s_c \in \mathbb{R}^C$ and a 2D spatial attention map $s_s \in \mathbb{R}^{H\times W}$. The formulation of the channel attention sub-module is similar to that of an SE block, except that it adopts more than one type of pooling operation to aggregate global information. In detail, it has two parallel branches using max-pool and avg-pool operations:

$$F_{\text{avg}}^c = \text{GAP}^s(X) \tag{80}$$

$$F_{\text{max}}^c = \text{GMP}^s(X) \tag{81}$$

$$s_c = \sigma(W_2\delta(W_1F_{\text{avg}}^c) + W_2\delta(W_1F_{\text{max}}^c)) \tag{82}$$

$$M_c(X) = s_cX \tag{83}$$

where GAP^s and GMP^s denote global average pooling and global max pooling operations in the spatial domain. The spatial attention sub-module models the spatial relationships of features, and is complementary to channel attention. Unlike channel attention, it applies a convolution layer with a large kernel to generate the attention map

$$F_{\text{avg}}^s = \text{GAP}^c(X) \tag{84}$$

$$F_{\text{max}}^s = \text{GMP}^c(X) \tag{85}$$

$$s_s = \sigma(\text{Conv}([F_{\text{avg}}^s; F_{\text{max}}^s])) \tag{86}$$

$$M_s(X) = s_sX \tag{87}$$

where $\text{Conv}(\cdot)$ represents a convolution operation, while GAP^c and GMP^c are global pooling operations in the channel domain. $[]$ denotes concatenation over channels. The overall attention process can be summarized as

$$X' = M_c(X) \tag{88}$$

$$Y = M_s(X') \tag{89}$$

Combining channel attention and spatial attention sequentially, CBAM can utilize both spatial and cross-channel relationships of features to tell the network *what* to focus on and *where* to focus. To be more specific, it emphasizes useful channels as well as enhancing informative local regions. Due to its lightweight design, CBAM can be integrated into any CNN architecture seamlessly with negligible additional cost. Nevertheless, there is still room for improvement in the channel & spatial attention mechanism. For instance, CBAM adopts a convolution to produce the spatial attention map, so the spatial sub-module may suffer from a limited receptive field.

3.6.3 BAM

At the same time as CBAM, Park et al. [117] proposed the *bottleneck attention module* (BAM), aiming to efficiently improve the representational capability of networks. It uses dilated convolution to enlarge the receptive field of the spatial attention sub-module, and build a *bottleneck structure* as suggested by ResNet to save computational cost.

For a given input feature map X, BAM infers the channel attention $s_c \in \mathbb{R}^C$ and spatial attention $s_s \in \mathbb{R}^{H\times W}$ in two parallel streams, then sums the two attention maps after resizing both branch outputs to $\mathbb{R}^{C\times H\times W}$. The channel attention branch, like an SE block, applies global average pooling to the feature map to aggregate global information, and then uses an MLP with channel dimensionality reduction. In order to utilize contextual information effectively, the

TABLE 7
Representative channel & spatial attention mechanisms sorted by date. Cls = classification, ICap = image captioning, Det = detection, Seg = segmentation, ISeg = instance segmentation, KP = keypoint detection, ReID = re-identification.$g(x)$ and $f(g(x), x)$ are the attention process described by Eq. 1. Ranges means the ranges of attention map. S or H means soft or hard attention. (A) element-wise product. (B) aggregate information via attention map.(I) focus the network on the discriminative region, (II) emphasize important channels, (III) capture long-range information, (IV) capture cross-domain interaction between any two domains.

Category	Method	Publication	Tasks	$g(x)$	$f(g(x), x)$	Ranges	S or H	Goals
Jointly predict channel & spatial attention map	Residual Attention [119]	CVPR2017	Cls	top-down network -> bottom down network -> 1×1 Convs -> Sigmoid	(A)	(0,1)	S	(I), (II)
	SCNet [120]	CVPR2020	Cls, Det, ISeg, KP	top-down network -> bottom down network -> identity add -> sigmoid	(A)	(0,1)	S	(II), (III)
	Strip Pooling [124]	CVPR2020	Seg	a)horizontal/vertical global pooling -> 1D Conv -> point-wise summation -> 1×1 Conv -> Sigmoid	(A)	(0,1)	S	(I), (II), (III)
Separately predict channel & spatial attention maps	SCA-CNN [50]	CVPR2017	ICap	a)spatial: fuse hidden state -> 1×1 Conv -> Softmax, b)channel: global average pooling -> MLP -> Softmax	(A)	(0,1)	S	(I), (II), (III)
	CBAM [6]	ECCV2018	Cls, Det	a)spatial: global pooling in channel dimension-> Conv -> Sigmoid, b)channel: global pooling in spatial dimension -> MLP -> Sigmoid	(A)	(0,1)	S	(I), (II), (III)
	BAM [6]	BMVC2018	Cls, Det	a)spatial: dilated Convs, b)channel: global average pooling -> MLP, c)fuse two branches	(A)	(0,1)	S	(I), (II), (III)
	scSE [123]	TMI2018	Seg	a)spatial: 1×1 Conv -> Sigmoid, b)channel: global average pooling -> MLP -> Sigmoid, c)fuse two branches	(A)	(0,1)	S	(I), (II), (III)
	Dual Attention [10]	CVPR2019	Seg	a)spatial: self-attention in spatial dimension, b)channel: self-attention in channel dimension, c) fuse two branches	(B)	(0,1)	S	(I), (II), (III)
	RGA [101]	CVPR2020	ReID	use self-attention to capture pairwise relations -> compute attention maps with the input and relation vectors	(A)	(0,1)	S	(I), (II), (III)
	Triplet Attention [121]	WACV2021	Cls, Det	compute attention maps for pairs of domains -> fuse different branches	(A)	(0,1)	S	(I), (IV)

spatial attention branch combines a bottleneck structure and dilated convolutions. Overall, BAM can be written as

$$s_c = \mathrm{BN}(W_2(W_1\mathrm{GAP}(X) + b_1) + b_2) \tag{90}$$
$$s_s = \mathrm{BN}(\mathrm{Conv}_2^{1\times1}(\mathrm{DC}_2^{3\times3}(\mathrm{DC}_1^{3\times3}(\mathrm{Conv}_1^{1\times1}(X))))) \tag{91}$$
$$s = \sigma(\mathrm{Expand}(s_s) + \mathrm{Expand}(s_c)) \tag{92}$$
$$Y = sX + X \tag{93}$$

where W_i, b_i denote weights and biases of fully connected layers respectively, $\mathrm{Conv}_1^{1\times1}$ and $\mathrm{Conv}_2^{1\times1}$ are convolution layers used for channel reduction. $\mathrm{DC}_i^{3\times3}$ denotes a dilated convolution with 3×3 kernel, applied to utilize contextual information effectively. Expand expands the attention maps s_s and s_c to $\mathbb{R}^{C\times H\times W}$.

BAM can emphasize or suppress features in both spatial and channel dimensions, as well as improving the representational power. Dimensional reduction applied to both channel and spatial attention branches enables it to be integrated with any convolutional neural network with little extra computational cost. However, although dilated convolutions enlarge the receptive field effectively, it still fails to capture long-range contextual information as well as encoding cross-domain relationships.

3.6.4 scSE

To aggregate global spatial information, an SE block applies global pooling to the feature map. However, it ignores pixel-wise spatial information, which is important in dense prediction tasks. Therefore, Roy et al. [123] proposed *spatial and channel SE blocks* (scSE). Like BAM, spatial SE blocks are used, complementing SE blocks, to provide spatial attention weights to focus on important regions.

Given the input feature map X, two parallel modules, spatial SE and channel SE, are applied to feature maps to encode spatial and channel information respectively. The channel SE module is an ordinary SE block, while the spatial SE module adopts 1×1 convolution for spatial squeezing. The outputs from the two modules are fused. The overall process can be written as

$$s_c = \sigma(W_2\delta(W_1\text{GAP}(X))) \tag{94}$$
$$X_{\text{chn}} = s_c X \tag{95}$$
$$s_s = \sigma(\text{Conv}^{1\times1}(X)) \tag{96}$$
$$X_{\text{spa}} = s_s X \tag{97}$$
$$Y = f(X_{\text{spa}}, X_{\text{chn}}) \tag{98}$$

where f denotes the fusion function, which can be maximum, addition, multiplication or concatenation.

The proposed scSE block combines channel and spatial attention to enhance features as well as capturing pixel-wise spatial information. Segmentation tasks are greatly benefited as a result. The integration of an scSE block in F-CNNs makes a consistent improvement in semantic segmentation at negligible extra cost.

3.6.5 Triplet Attention

In CBAM and BAM, channel attention and spatial attention are computed independently, ignoring relationships between these two domains [121]. Motivated by spatial attention, Misra et al. [121] proposed *triplet attention*, a lightweight but effective attention mechanism to capture cross-domain interaction.

Given an input feature map X, triplet attention uses three branches, each of which plays a role in capturing cross-domain interaction between any two domains from H, W and C. In each branch, rotation operations along different axes are applied to the input first, and then a Z-pool layer is responsible for aggregating information in the zeroth dimension. Finally, a standard convolution layer with kernel size $k \times k$ models the relationship between the last two domains. This process can be written as

$$X_1 = \text{Pm}_1(X) \tag{99}$$
$$X_2 = \text{Pm}_2(X) \tag{100}$$
$$s_0 = \sigma(\text{Conv}_0(\text{Z-Pool}(X))) \tag{101}$$
$$s_1 = \sigma(\text{Conv}_1(\text{Z-Pool}(X_1))) \tag{102}$$
$$s_2 = \sigma(\text{Conv}_2(\text{Z-Pool}(X_2))) \tag{103}$$
$$Y = \frac{1}{3}(s_0 X + \text{Pm}_1^{-1}(s_1 X_1) + \text{Pm}_2^{-1}(s_2 X_2)) \tag{104}$$

where Pm_1 and Pm_2 denote rotation through 90° anti-clockwise about the H and W axes respectively, while Pm_i^{-1} denotes the inverse. Z-Pool concatenates max-pooling and average pooling along the zeroth dimension.

$$Y = \text{Z-Pool}(X) = [\text{GMP}(X); \text{GAP}(X)] \tag{105}$$

Unlike CBAM and BAM, triplet attention stresses the importance of capturing cross-domain interactions instead of computing spatial attention and channel attention independently. This helps to capture rich discriminative feature representations. Due to its simple but efficient structure, triplet attention can be easily added to classical backbone networks.

3.6.6 SimAM

Yang et al. [118] also stress the importance of learning attention weights that vary across both channel and spatial domains in proposing SimAM, a simple, parameter-free attention module capable of directly estimating 3D weights instead of expanding 1D or 2D weights. The design of SimAM is based on well-known neuroscience theory, thus avoiding need for manual fine tuning of the network structure.

Motivated by the spatial suppression phenomenon [176], they propose that a neuron which shows suppression effects should be emphasized and define an energy function for each neuron as:

$$e_t(w_t, b_t, y, x_i) = (y_t - \hat{t})^2 + \frac{1}{M-1}\sum_{i=1}^{M-1}(y_o - \hat{x}_i) \tag{106}$$

where $\hat{t} = w_t t + b_t$, $\hat{x}_i = w_t x_i + b_t$, and t and x_i are the target unit and all other units in the same channel; $i \in 1, \ldots, N$, and $N = H \times W$.

An optimal closed-form solution for Eq. 106 exists:

$$e_t^* = \frac{4(\hat{\sigma}^2 + \lambda)}{(t - \hat{\mu})^2 + 2\hat{\sigma}^2 + 2\lambda} \tag{107}$$

where $\hat{\mu}$ is the mean of the input feature and $\hat{\sigma}^2$ is its variance. A sigmoid function is used to control the output range of the attention vector; an element-product is applied to get the final output:

$$Y = \text{Sigmoid}\left(\frac{1}{E}\right) X \tag{108}$$

This work simplifies the process of designing attention and successfully proposes a novel 3-D weight parameter-free attention module based on mathematics and neuroscience theories.

3.6.7 Coordinate attention

An SE block aggregates global spatial information using global pooling before modeling cross-channel relationships, but neglects the importance of positional information. BAM and CBAM adopt convolutions to capture local relations, but fail to model long-range dependencies. To solve these problems, Hou et al. [129] proposed *coordinate attention*, a novel attention mechanism which embeds positional information into channel attention, so that the network can focus on large important regions at little computational cost.

The coordinate attention mechanism has two consecutive steps, coordinate information embedding and coordinate attention generation. First, two spatial extents of pooling kernels encode each channel horizontally and vertically. In the second step, a shared 1×1 convolutional transformation function is applied to the concatenated outputs of the two pooling layers. Then coordinate attention splits the resulting tensor into two separate tensors to yield attention vectors